

Laboratory Management System - ERD

Schema overview and detail-page index

All 25 tables are listed. Only selected transaction relationships are shown; this is not the full ERD.

Accounts, Classes and Q&A

A1

STUDENT
USER_ACCOUNT
CLASS_GROUP
GROUP_MEMBER
CHAT_EXCHANGE
KNOWLEDGE_ARTICLE

Reservations and Approvals

A2

RESERVATION
REQUEST_REVISION
APPROVAL
REQUEST_ITEM
REQUEST_MEMBER

Inventory and Disposal

A3

ITEM_CATEGORY
ITEM
STOCK_MOVEMENT
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Borrowing and Returns

A4

BORROWING
BORROWING_ITEM
BORROWING_MEMBER
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Laboratories and Schedules

A5

TERM
SCHEDULE_BLOCK
LABORATORY

Usage, Daily Tasks and Clearance

A6

USAGE_LOG
DAILY_TASK
CLEARANCE

Selected transaction relationships - exact FK pairs

ID	Parent / PK	Cardinality	Child / FK	Details
R13	RESERVATION PK request_id	1 0..*	REQUEST_REVISION FK request_id	A2 / A2
R17	REQUEST_REVISION PK revision_id	1 0..1	APPROVAL FK revision_id	A2 / A2
R23	REQUEST_REVISION PK revision_id	1 0..*	BORROWING FK revision_id	A2 / A4
R25	BORROWING PK borrowing_id	1 0..*	BORROWING_ITEM FK borrowing_id	A4 / A4
R29	BORROWING_ITEM PK borrowing_item_id	1 0..*	RETURN_ENTRY FK borrowing_item_id	A4 / A4
R45	REQUEST_REVISION PK revision_id	1 0..*	USAGE_LOG FK revision_id	A2 / A6

All remaining relationships and complete attributes are shown in Details A1-A6.

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Detail A1 - Accounts, Classes and Q&A

Left: parents per child. Right: children per parent. ? = nullable. R-tags match the FK rows.

STUDENT		D1
PK	student_id	
UK	student_number	
	full_name	
	program	
	year_section	

USER_ACCOUNT		D1
PK	account_id	
FK/UK	student_id ?	R01
FK	staff_lab_id ?	R02
UK	login_identifier	
	password_hash	
	role	
	status	
	last_login_at ?	

CLASS_GROUP		D1
PK	group_id	
FK	term_id	R03
FK	faculty_id	R04
FK	representative_id	R05
	subject_code	
	year_section	
	active	

GROUP_MEMBER		D1
PK	group_member_id	
FK	group_id	R06
FK	student_id	R07
	active	

CHAT_EXCHANGE		D9
PK	chat_id	
FK	requester_id	R51
	question	
	answer	
	intent	
	response_status	
	asked_at	

KNOWLEDGE_ARTICLE		D8
PK	article_id	
	topic	
	content	
	category	
	updated_at	

ID	Parent / PK	Cardinality	Child / FK	Details
R01	STUDENT PK student_id	0..1 0..1	USER_ACCOUNT FK student_id	A1 / A1
R02	LABORATORY PK lab_id	0..1 0..*	USER_ACCOUNT FK staff_lab_id	A5 / A1
R03	TERM PK term_id	1 0..*	CLASS_GROUP FK term_id	A5 / A1
R04	USER_ACCOUNT PK account_id	1 0..*	CLASS_GROUP FK faculty_id	A1 / A1
R05	USER_ACCOUNT PK account_id	1 0..*	CLASS_GROUP FK representative_id	A1 / A1
R06	CLASS_GROUP PK group_id	1 0..*	GROUP_MEMBER FK group_id	A1 / A1
R07	STUDENT PK student_id	1 0..*	GROUP_MEMBER FK student_id	A1 / A1
R51	USER_ACCOUNT PK account_id	1 0..*	CHAT_EXCHANGE FK requester_id	A1 / A1

KNOWLEDGE_ARTICLE has no declared FK. Ordinary students do not need login accounts.

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Detail A2 - Reservations and Approvals

Left: parents per child. Right: children per parent. ? = nullable. R-tags match the FK rows.

RESERVATION	D2	APPROVAL	D2
PK request_id		PK approval_id	
FK requester_id	R11	FK/UK revision_id	R17
FK group_id	R12	FK approver_id	R18
reservation_type ?		route_order	
UK reference_no		decision	
purpose		decided_at ?	
status			
filed_at			

REQUEST_REVISION	D2	REQUEST_ITEM	D2
PK revision_id		PK request_item_id	
FK request_id	R13	FK revision_id	R19
FK lab_id	R14	FK item_id	R20
FK block_id ?	R15	qty_requested	
FK changed_by	R16		
revision_no			
starts_at			
ends_at			
request_basis			
usage_type			
is_current			

REQUEST_MEMBER	D2
PK request_member_id	
FK revision_id	R21
FK student_id	R22

ID	Parent / PK	Cardinality	Child / FK	Details
R11	USER_ACCOUNT PK account_id	1 0..*	RESERVATION FK requester_id	A1 / A2
R12	CLASS_GROUP PK group_id	1 0..*	RESERVATION FK group_id	A1 / A2
R13	RESERVATION PK request_id	1 0..*	REQUEST_REVISION FK request_id	A2 / A2
R14	LABORATORY PK lab_id	1 0..*	REQUEST_REVISION FK lab_id	A5 / A2
R15	SCHEDULE_BLOCK PK block_id	0..1 0..*	REQUEST_REVISION FK block_id	A5 / A2
R16	USER_ACCOUNT PK account_id	1 0..*	REQUEST_REVISION FK changed_by	A1 / A2
R17	REQUEST_REVISION PK revision_id	1 0..1	APPROVAL FK revision_id	A2 / A2
R18	USER_ACCOUNT PK account_id	1 0..*	APPROVAL FK approver_id	A1 / A2
R19	REQUEST_REVISION PK revision_id	1 0..*	REQUEST_ITEM FK revision_id	A2 / A2
R20	ITEM PK item_id	1 0..*	REQUEST_ITEM FK item_id	A3 / A2
R21	REQUEST_REVISION PK revision_id	1 0..*	REQUEST_MEMBER FK revision_id	A2 / A2
R22	STUDENT PK student_id	1 0..*	REQUEST_MEMBER FK student_id	A1 / A2

Each revision has zero or one approval. The selected Faculty or Dean makes the final decision.

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Detail A3 - Inventory and Disposal

Left: parents per child. Right: children per parent. ? = nullable. R-tags match the FK rows.

ITEM_CATEGORY		D4
PK	category_id	
UK	category_name	
	description	

ITEM		D4
PK	item_id	
FK	lab_id	R35
FK	category_id	R36
	item_name	
	item_type	
	unit	
	reorder_level	
	condition_status	

STOCK_MOVEMENT		D4
PK	movement_id	
FK	item_id	R37
FK	recorded_by	R38
FK	borrowing_item_id ?	R39
FK	return_entry_id ?	R40
FK	disposal_id ?	R41
	on_hand_delta	
	owned_delta	
	movement_type	
	recorded_at	

DISPOSAL		D10
PK	disposal_id	
FK	item_id	R42
FK	recorded_by	R43
FK	return_entry_id ?	R44
	quantity	
	waste_class	
	reason	
	disposed_at	
	status	

ID	Parent / PK	Cardinality	Child / FK	Details
R35	LABORATORY PK lab_id	1 0..*	ITEM FK lab_id	A5 / A3
R36	ITEM_CATEGORY PK category_id	1 0..*	ITEM FK category_id	A3 / A3
R37	ITEM PK item_id	1 0..*	STOCK_MOVEMENT FK item_id	A3 / A3
R38	USER_ACCOUNT PK account_id	1 0..*	STOCK_MOVEMENT FK recorded_by	A1 / A3
R39	BORROWING_ITEM PK borrowing_item_id	0..1 0..*	STOCK_MOVEMENT FK borrowing_item_id	A4 / A3
R40	RETURN_ENTRY PK return_entry_id	0..1 0..*	STOCK_MOVEMENT FK return_entry_id	A4 / A3
R41	DISPOSAL PK disposal_id	0..1 0..*	STOCK_MOVEMENT FK disposal_id	A3 / A3
R42	ITEM PK item_id	1 0..*	DISPOSAL FK item_id	A3 / A3
R43	USER_ACCOUNT PK account_id	1 0..*	DISPOSAL FK recorded_by	A1 / A3
R44	RETURN_ENTRY PK return_entry_id	0..1 0..*	DISPOSAL FK return_entry_id	A4 / A3

Stock movements retain their issue, return or disposal source. Lost items are not physical waste.

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Detail A4 - Borrowing and Returns

Left: parents per child. Right: children per parent. ? = nullable. R-tags match the FK rows.

BORROWING		D5
PK	borrowing_id	
FK	revision_id	R23
FK	issued_by	R24
UK	slip_no	
	issued_at	
	status	

BORROWING_MEMBER		D5
PK	borrow_member_id	
FK	borrowing_id	R27
FK	request_member_id	R28

BORROWING_ITEM		D5
PK	borrowing_item_id	
FK	borrowing_id	R25
FK	request_item_id	R26
	qty_issued	

RETURN_ENTRY		D5
PK	return_entry_id	
FK	borrowing_item_id	R29
FK	received_by	R30
	qty_good	
	qty_broken	
	qty_lost	
	qty_consumed	
	recorded_at	

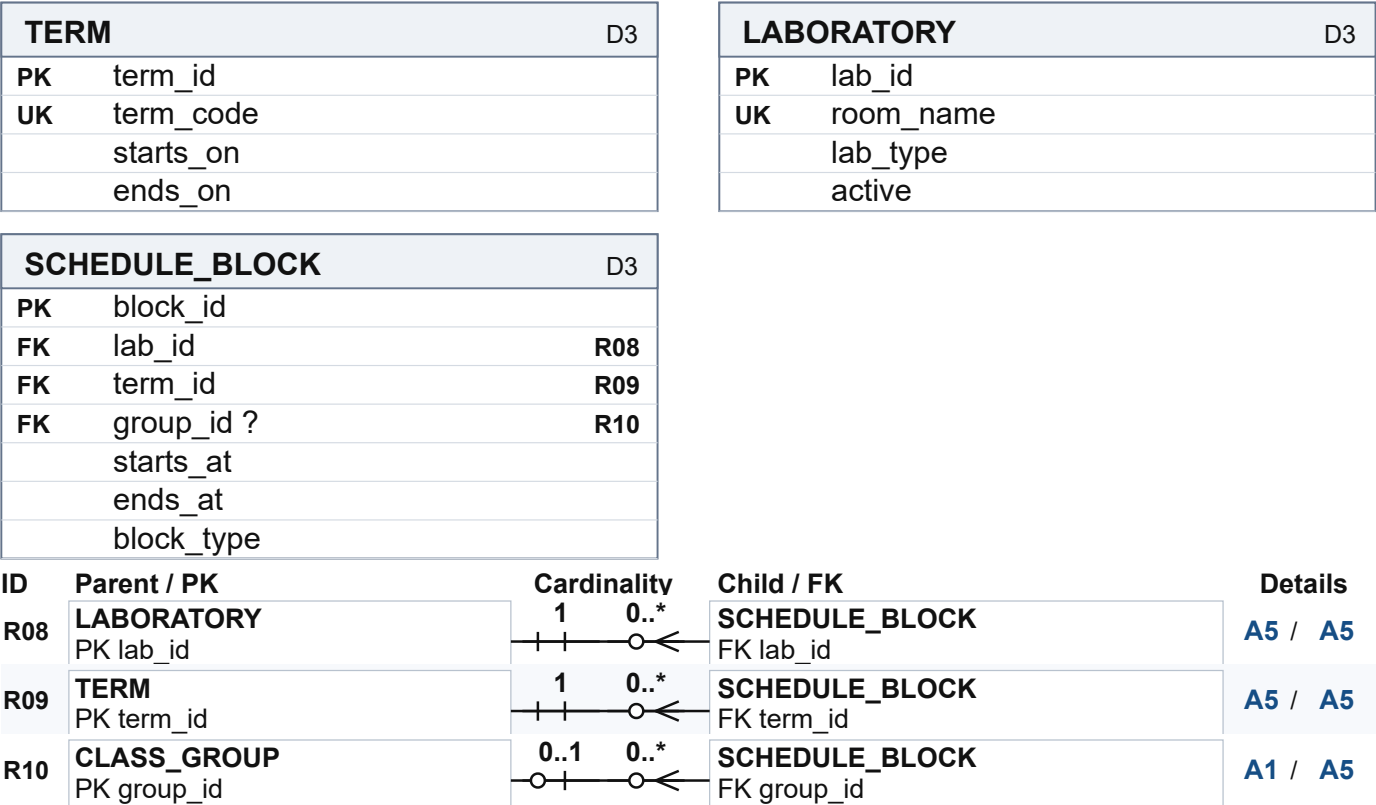
ID	Parent / PK	Cardinality	Child / FK	Details
R23	REQUEST_REVISION PK revision_id	1 0..*	BORROWING FK revision_id	A2 / A4
R24	USER_ACCOUNT PK account_id	1 0..*	BORROWING FK issued_by	A1 / A4
R25	BORROWING PK borrowing_id	1 0..*	BORROWING_ITEM FK borrowing_id	A4 / A4
R26	REQUEST_ITEM PK request_item_id	1 0..*	BORROWING_ITEM FK request_item_id	A2 / A4
R27	BORROWING PK borrowing_id	1 0..*	BORROWING_MEMBER FK borrowing_id	A4 / A4
R28	REQUEST_MEMBER PK request_member_id	1 0..*	BORROWING_MEMBER FK request_member_id	A2 / A4
R29	BORROWING_ITEM PK borrowing_item_id	1 0..*	RETURN_ENTRY FK borrowing_item_id	A4 / A4
R30	USER_ACCOUNT PK account_id	1 0..*	RETURN_ENTRY FK received_by	A1 / A4

Requested and issued quantities remain separate. Return entries preserve partial-return history.

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Detail A5 - Laboratories and Schedules

Left: parents per child. Right: children per parent. ? = nullable. R-tags match the FK rows.



Schedule blocks contain planned or vacant time slots, not completed usage.

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Detail A6 - Usage, Daily Tasks and Clearance

Left: parents per child. Right: children per parent. ? = nullable. R-tags match the FK rows.

USAGE_LOG			D11
PK	usage_id		
FK	revision_id		R45
FK	borrowing_id ?		R46
FK	faculty_id		R47
FK	recorded_by		R48
	actual_start		
	actual_end		
	remarks ?		

DAILY_TASK			D7
PK	task_id		
FK	lab_id		R49
FK	recorded_by		R50
	task_date		
	activity		
	status		
	work_availability		
	overtime_hours		
	remarks ?		

CLEARANCE			D6
PK	clearance_id		
FK	return_entry_id		R31
FK	borrow_member_id		R32
FK	raised_by		R33
FK	settled_by ?		R34
	quantity		
	reason		
	status		
	raised_at		
	settled_at ?		

ID	Parent / PK	Cardinality	Child / FK	Details
R31	RETURN_ENTRY PK return_entry_id	1 0..*	CLEARANCE FK return_entry_id	A4 / A6
R32	BORROWING_MEMBER PK borrow_member_id	1 0..*	CLEARANCE FK borrow_member_id	A4 / A6
R33	USER_ACCOUNT PK account_id	1 0..*	CLEARANCE FK raised_by	A1 / A6
R34	USER_ACCOUNT PK account_id	0..1 0..*	CLEARANCE FK settled_by	A1 / A6
R45	REQUEST_REVISION PK revision_id	1 0..*	USAGE_LOG FK revision_id	A2 / A6
R46	BORROWING PK borrowing_id	0..1 0..*	USAGE_LOG FK borrowing_id	A4 / A6
R47	USER_ACCOUNT PK account_id	1 0..*	USAGE_LOG FK faculty_id	A1 / A6
R48	USER_ACCOUNT PK account_id	1 0..*	USAGE_LOG FK recorded_by	A1 / A6
R49	LABORATORY PK lab_id	1 0..*	DAILY_TASK FK lab_id	A5 / A6
R50	USER_ACCOUNT PK account_id	1 0..*	DAILY_TASK FK recorded_by	A1 / A6

Completed USAGE_LOG records supply the End-Term Report. Daily tasks and clearance remain separate.